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EE 456 RF & mm-Wave Active Circuits



Design Project 2

Low-Noise Amplifier Matching Network Design

Design Project #2

Element	Score	Max
Design		300
Input Matching Network		100
Output Matching Network		100
Simulations (MATLAB)		200
Simulations (ADS)		100
Comparison		100
Presentation		100
Overall		1000

Design Project Objective

Design a 15 GHz low noise amplifier using the MGF4941AL super-low noise InGaAs HEMT biased at $V_{DS} = 2$ V and $I_{DS} = 10$ mA . The amplifier must meet the following specifications:

$$\begin{array}{l} G_T \geq 12 \text{ dB} \quad \& \quad F \leq 0.6 \text{ dB} \\ VSWR_{\text{IMN}} \leq 1.5 \quad \& \quad VSWR_{\text{OMN}} \leq 1.5 \end{array}$$

Note that the noise parameters at 15 GHz for the MGF4941AL HEMT biased at $V_{DS} = 2$ V and $I_{DS} = 10$ mA are given by:

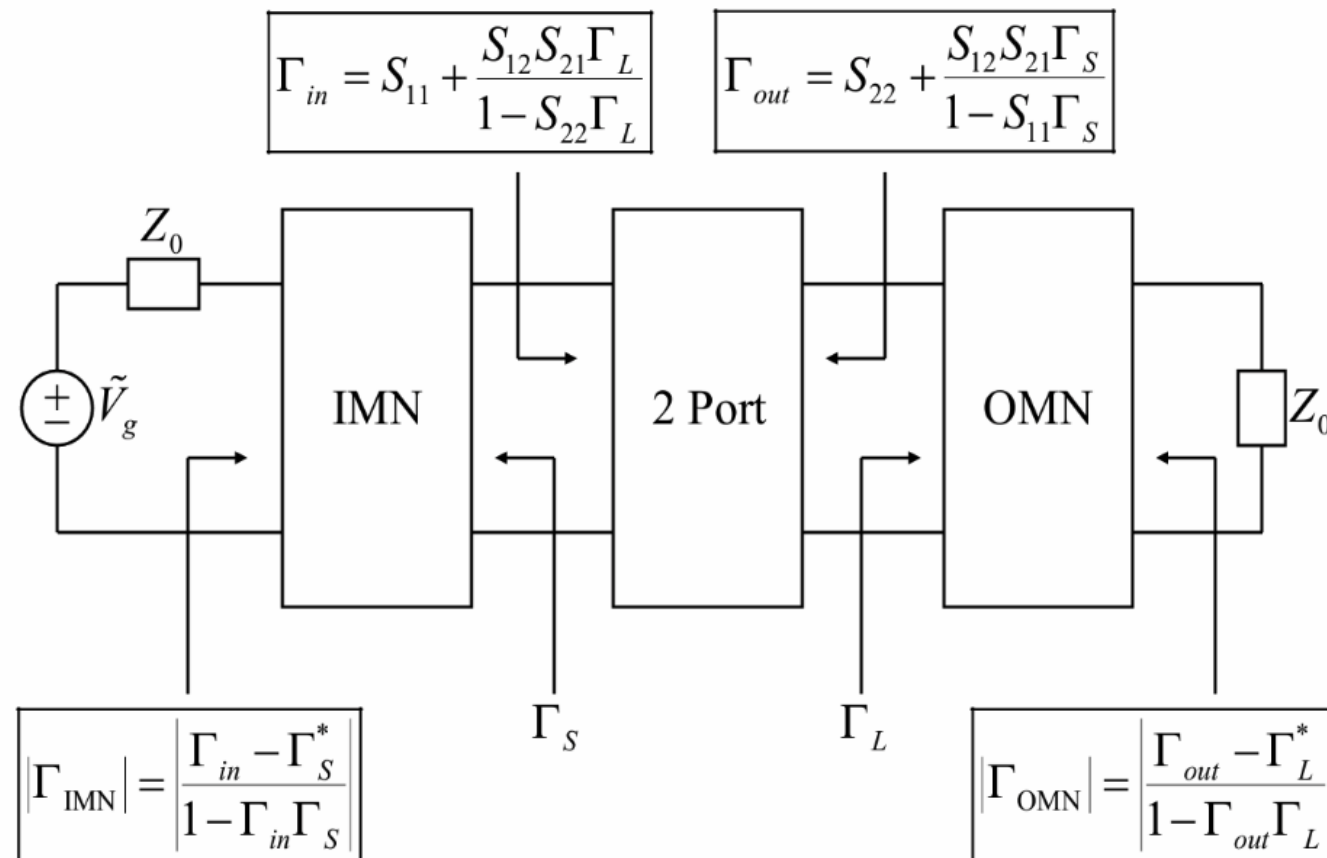
$$\begin{array}{l} F_{\text{min}} = 0.4450 \text{ dB} \\ \Gamma_{\text{OPT}} = 0.3240 \angle -165.1500^\circ \\ r_n = 0.0760 \end{array}$$

Design and Simulation Results

Table 1

	Design	Simulated Matlab	Simulated ADS	Units
Γ_S	(0.4256 , -136.0227°)	-----	-----	---
Γ_L	(0.2904 , +146.0865°)	-----	-----	---
Γ_{in}	(0.4742 , +123.6207°)	(0.4688 , -10.5628°)	(0.5638 , +112.7047°)	---
Γ_{out}	(0.3642 , -168.6686°)	(0.3555 , -25.0029°)	(0.3642 , -168.6684°)	---
$ \Gamma_{IMN} $	0.1350	0.3031	0.3031	---
$ \Gamma_{IMN} $	-17.3944	-10.2445	-10.2445	dB
$VSWR_{IMN}$	1.4983	1.8879	1.8879	---
$ \Gamma_{OMN} $	0.1630	0.1629	0.1629	---
$ \Gamma_{OMN} $	-15.7583	-15.7583	-15.7583	dB
$VSWR_{OMN}$	1.3894	1.3894	1.3894	---
G_T	12.2599	12.1483	12.1483	dB
F_{dB}	0.5803	0.5803	0.5803	dB

LNA Design: Network Layout / Reflection Coefficients



MAG Design



Design Parameter	Mag Design	Units
Γ_S	(0.7088 , -114.5883°)	---
Γ_L	(0.5871 , +152.4980°)	---
Γ_{in}	(0.7088 , +114.5883°)	---
Γ_{out}	(0.5871 , -152.4980°)	---
$ \Gamma_{IMN} $	0.0000	---
$ \Gamma_{IMN} $	-319.0497	dB
$VSWR_{IMN}$	1.0000	---
$ \Gamma_{OMN} $	0.0000	---
$ \Gamma_{OMN} $	314.4523	dB
$VSWR_{OMN}$	1.0000	---
G_T	12.9211	dB
F_{dB}	1.7920	dB

MAG Design

Design Parameter	Mag Design	Design Specs	Units	Design - Pass / Fail
Γ_S	$(0.7088 \angle + 114.5883^\circ)$	-----	----	-----
Γ_L	$(0.5871 \angle - 152.4980^\circ)$	-----	----	-----
Γ_{in}	$(0.7088 \angle + 114.5883^\circ)$	-----	----	-----
Γ_{out}	$(0.5871 \angle - 152.4980^\circ)$	-----	----	-----
$ \Gamma_{IMN} $	0.0000	-----	----	-----
$ \Gamma_{IMN} $	-319.0497	-----	dB	-----
$VSWR_{IMN}$	1.0000	≤ 1.5000	----	Pass
$ \Gamma_{OMN} $	0.0000	-----	----	-----
$ \Gamma_{OMN} $	314.4523	-----	dB	-----
$VSWR_{OMN}$	1.0000	≤ 1.5000	---	Fail
G_T	12.9211	≥ 12.0000	dB	Pass
F_{dB}	1.7920	≤ 0.6000	dB	Fail

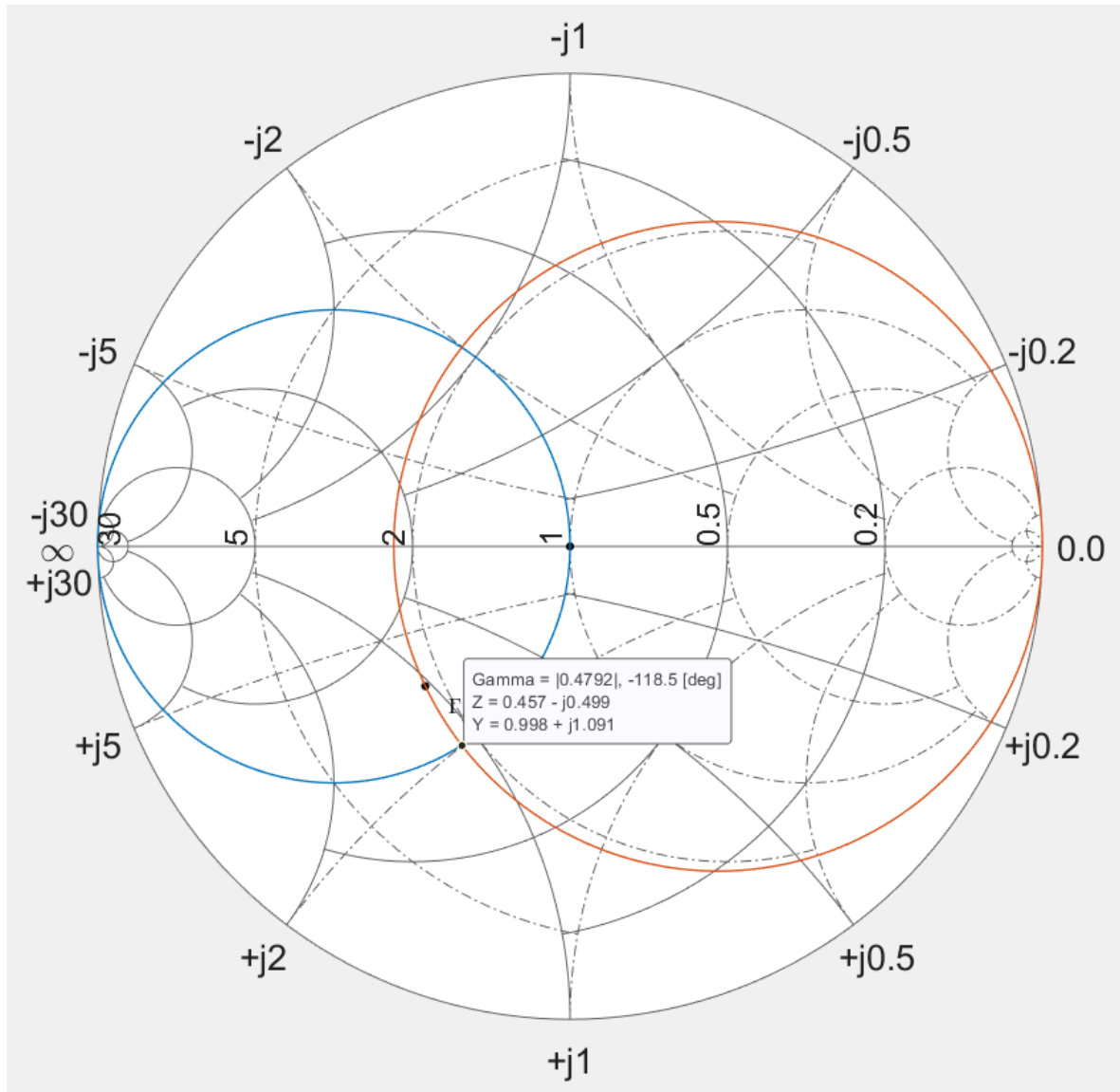
$F_{\min}$ Design

	$F_{\min}$ Design	Units
Γ_S	$0.3240 < -165.1500^\circ$	---
Γ_L	$0.2726 < -177.7155^\circ$	---
Γ_{in}	$0.5392 < +120.7417^\circ$	---
Γ_{out}	$0.2726 < +177.7155^\circ$	---
$ \Gamma_{IMN} $	0.4325	---
$ \Gamma_{IMN} $	-7.2798	dB
$VSWR_{IMN}$	2.5244	---
$ \Gamma_{OMN} $	0.0000	---
$ \Gamma_{OMN} $	-300.0000	dB
$VSWR_{OMN}$	1.0000	---
G_T	11.5032	dB
F_{dB}	0.4450	dB

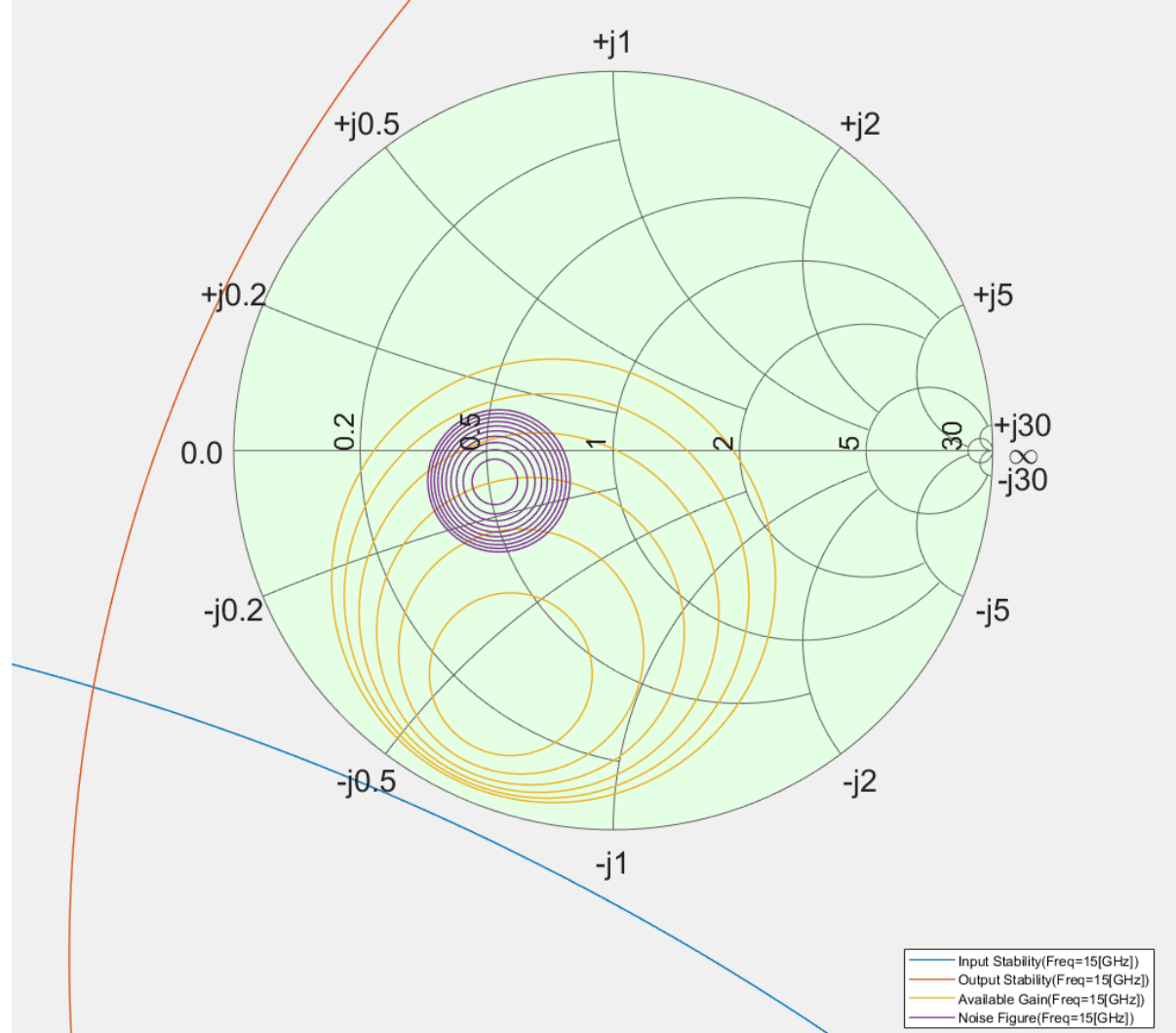
$F_{\min}$ Design



Design Parameter	$F_{\min}$ Design	Design Specs	Units	Design - Pass / Fail
Γ_S	(0.3240 , -165.1500°)	-----	----	-----
Γ_L	(0.2726 , -177.7155°)	-----	----	-----
Γ_{in}	(0.5392 , +120.7417°)	-----	----	-----
Γ_{out}	(0.2726 , +177.7155°)	-----	----	-----
$ \Gamma_{IMN} $	0.4325	-----	----	-----
$ \Gamma_{IMN} $	-7.2798	-----	dB	-----
$VSWR_{IMN}$	2.5244	≤ 1.5000	----	Fail
$ \Gamma_{OMN} $	0.0000	-----	----	-----
$ \Gamma_{OMN} $	-300.0000	-----	dB	-----
$VSWR_{OMN}$	1.0000	≤ 1.5000	---	Fail
G_T	11.5032	≥ 12.0000	dB	Fail
F_{dB}	0.4450	≤ 0.6000	dB	Pass



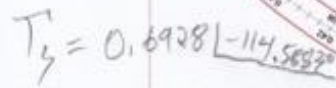
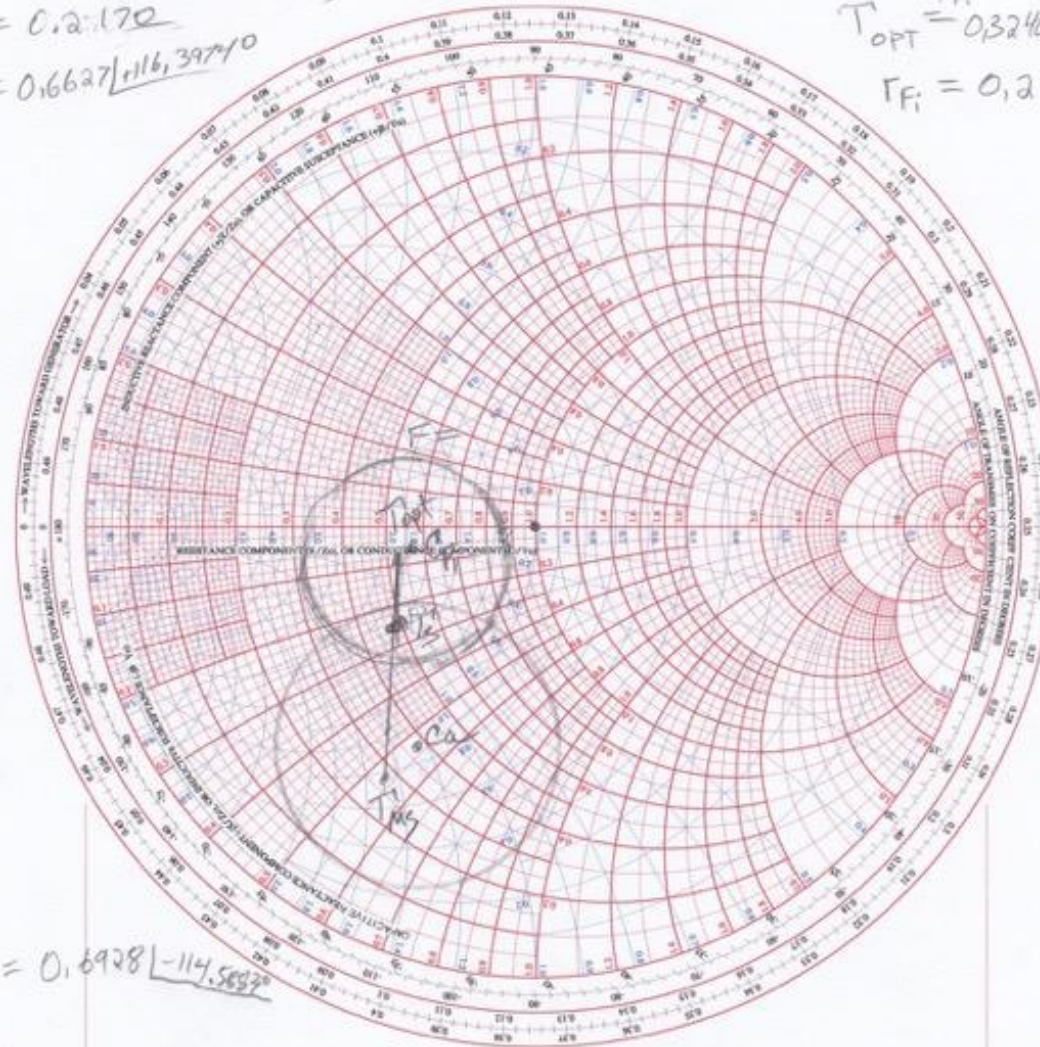
Constant Resistance Circle with Γ_s , its point of intersection with the constant conductance circle was used to determine the length of the transmission line between the amp and the stub in the Input Matching Network.

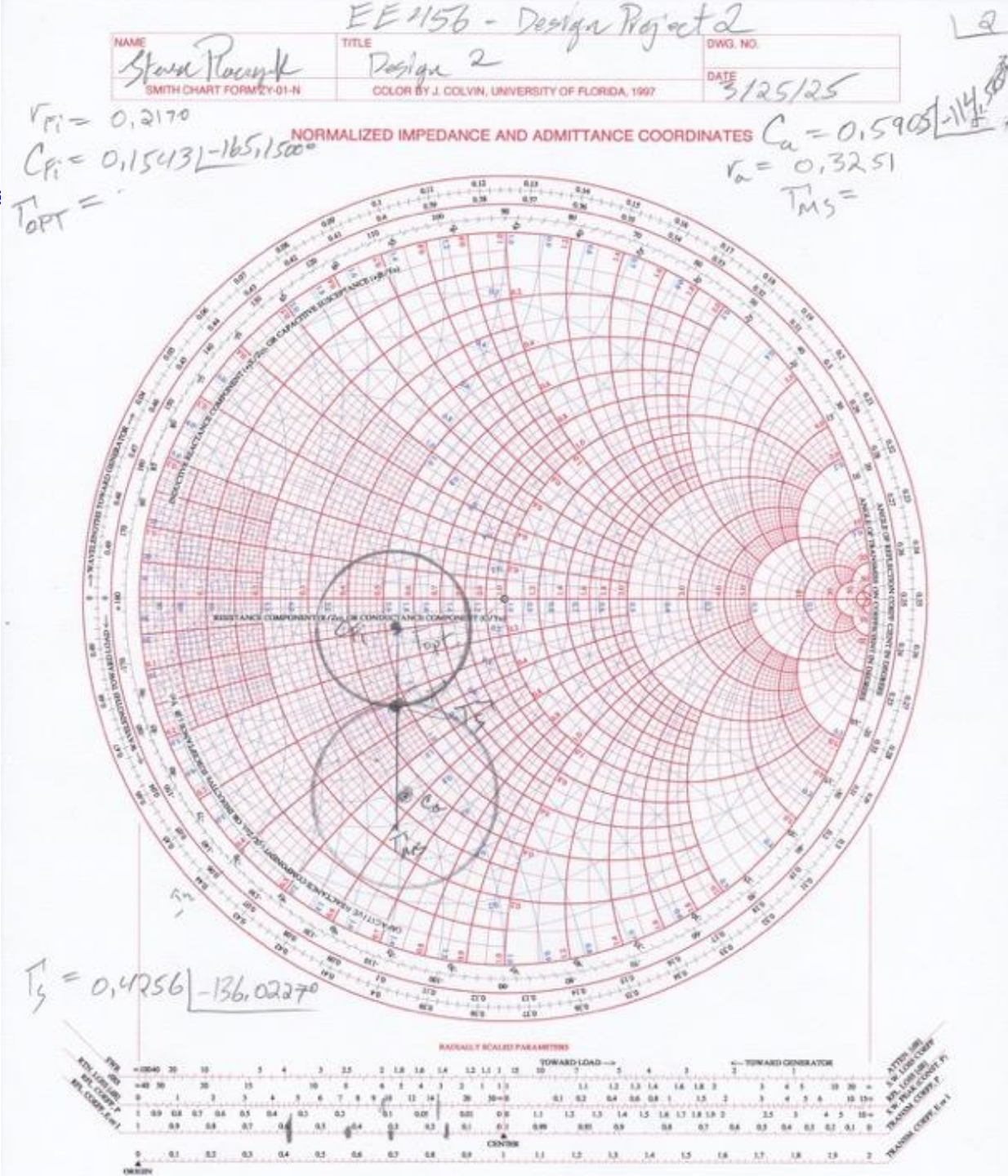
Un-Matched Amplifier Input/Output Stability, G_a , & Noise Figure Circles

$C_a = (0,6928 \mid -114,588^\circ)$
 $r_a = 0,2170$
 $T_{MS} = 0,6627 \mid +116,397^\circ$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$C_{F1} = (0,3011 \mid -165,552^\circ)$
 $T_{OPT} = 0,3240 \mid -165,550^\circ$
 $r_{F1} = 0,2170$



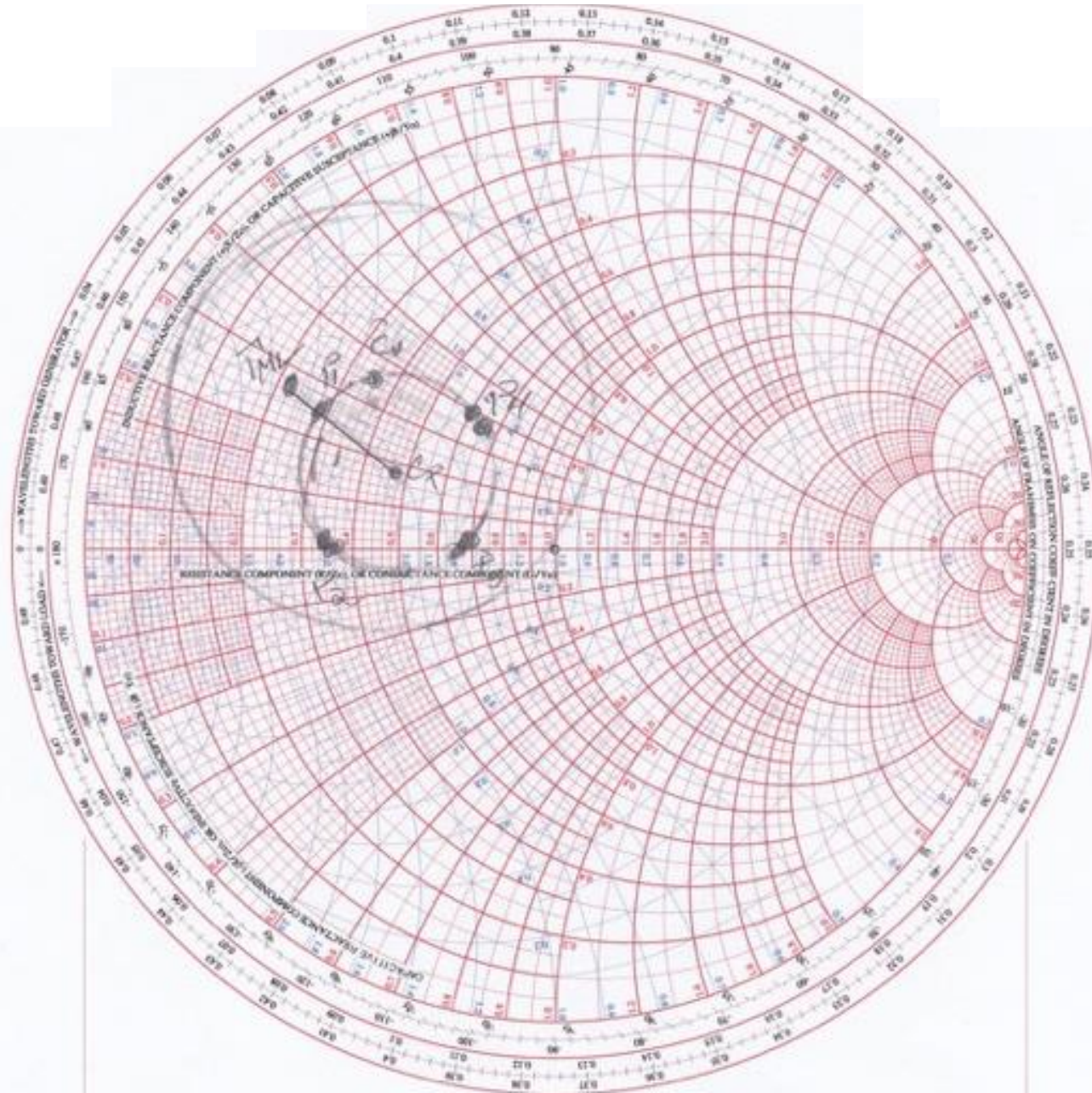


Design 1



Design Parameter	Design 1	Design Spec	Units	Design - Pass / Fail
Γ_S	(0.4264 , -135.9051°)	-----	----	-----
Γ_L	(0.4982, +157.5665°)	-----	----	-----
Γ_{in}	(0.6627, +116.3974°)	-----	----	-----
Γ_{out}	(0.4982 , -157.5655°)	-----	----	-----
$ \Gamma_{IMN} $	0.1173	-----	----	-----
$ \Gamma_{IMN} $	-18.6120 dB	-----	dB	-----
$VSWR_{IMN}$	1.2658	≤ 1.5000	----	Pass
$ \Gamma_{OMN} $	0.0000	-----	----	-----
$ \Gamma_{OMN} $	-300.0000	-----	dB	-----
$VSWR_{OMN}$	1.0000	≤ 1.5000	---	Pass
G_T	12.8123	≥ 12.0000	dB	Pass
F_{dB}	1.1433	≤ 0.6000	dB	FAIL

$$P_{3\Gamma_L} = 0.2904 \angle +146.0865^\circ$$



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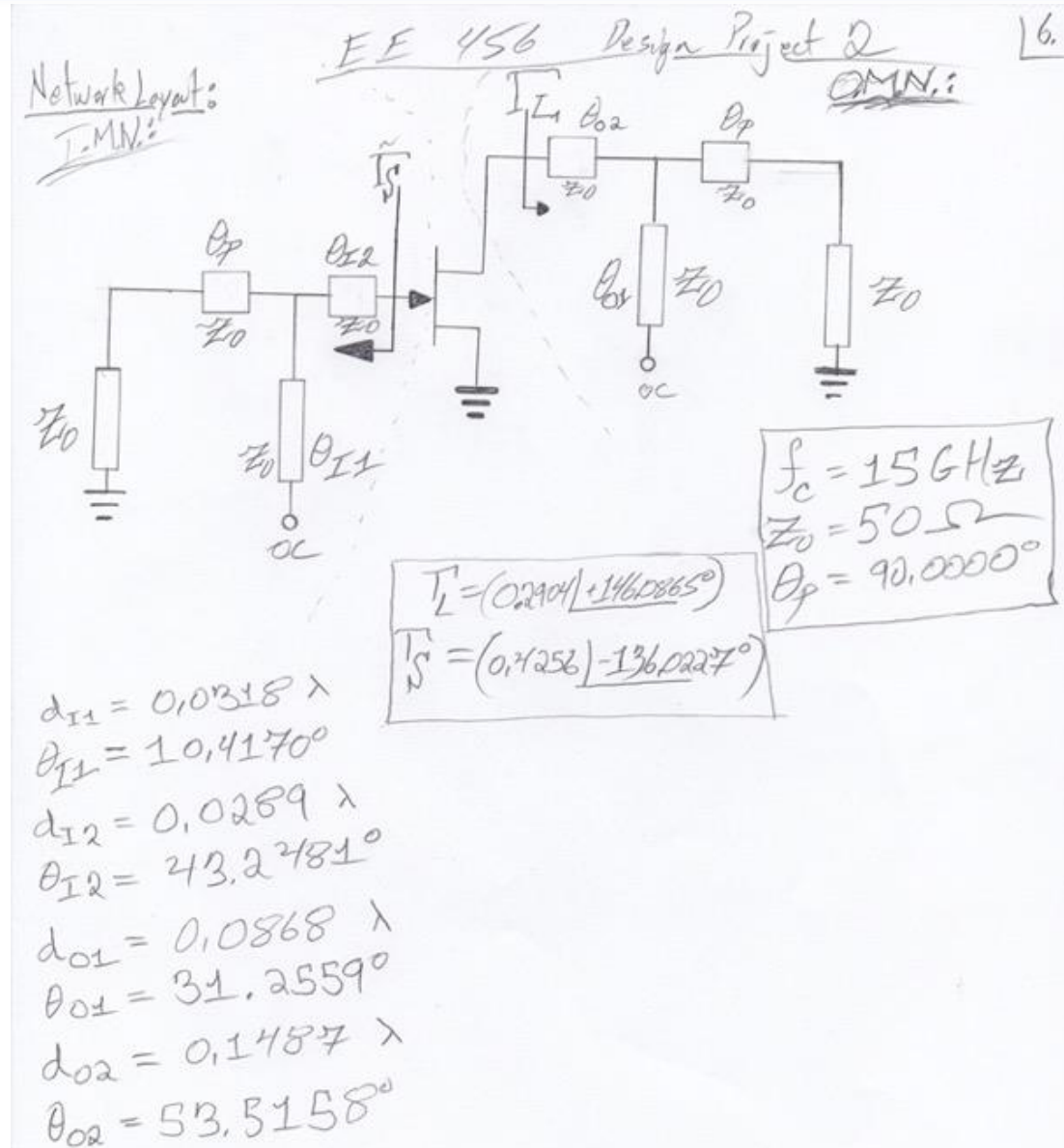
Design Project 2

Blank

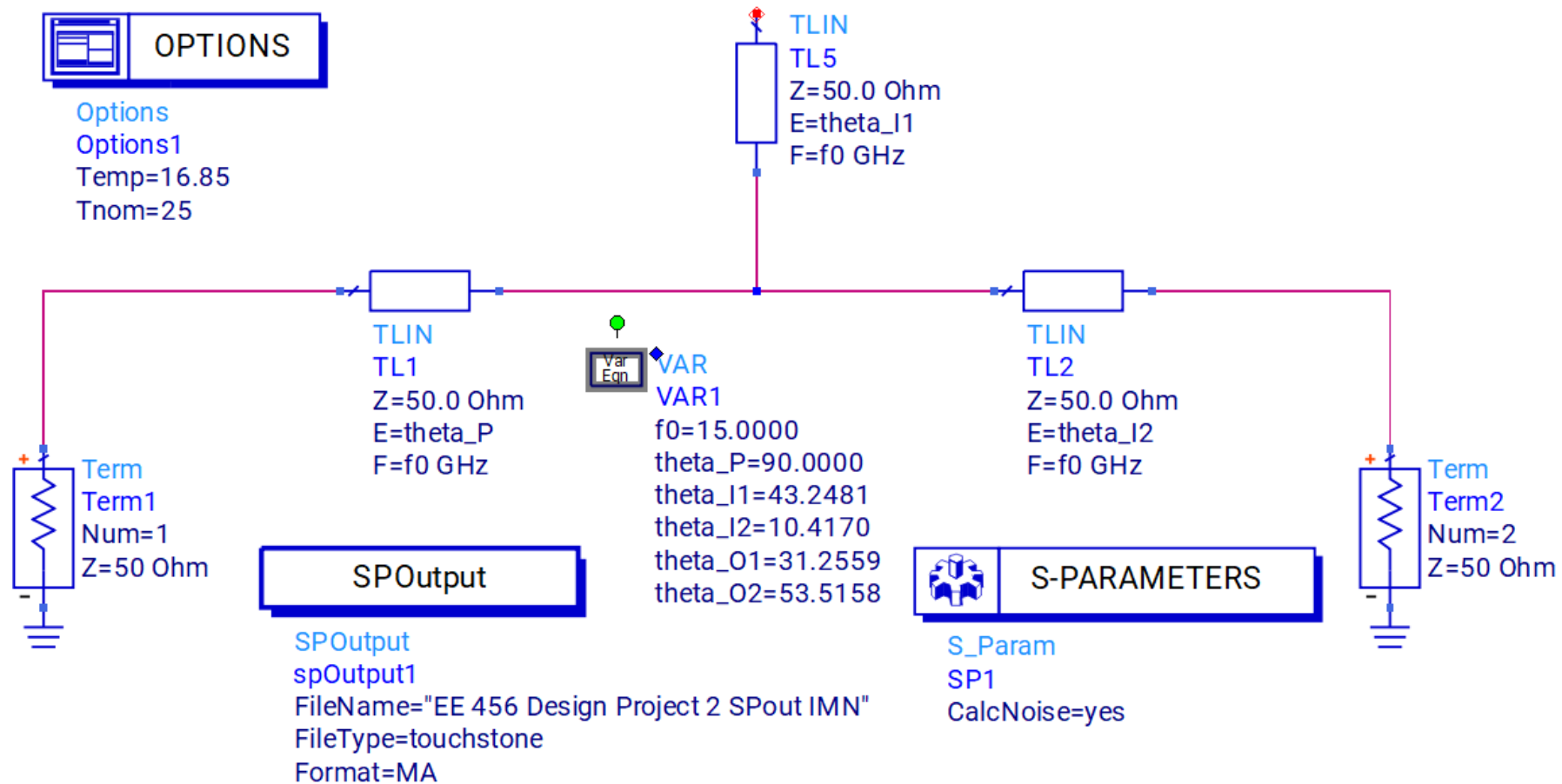
Γ_L - 4 Points

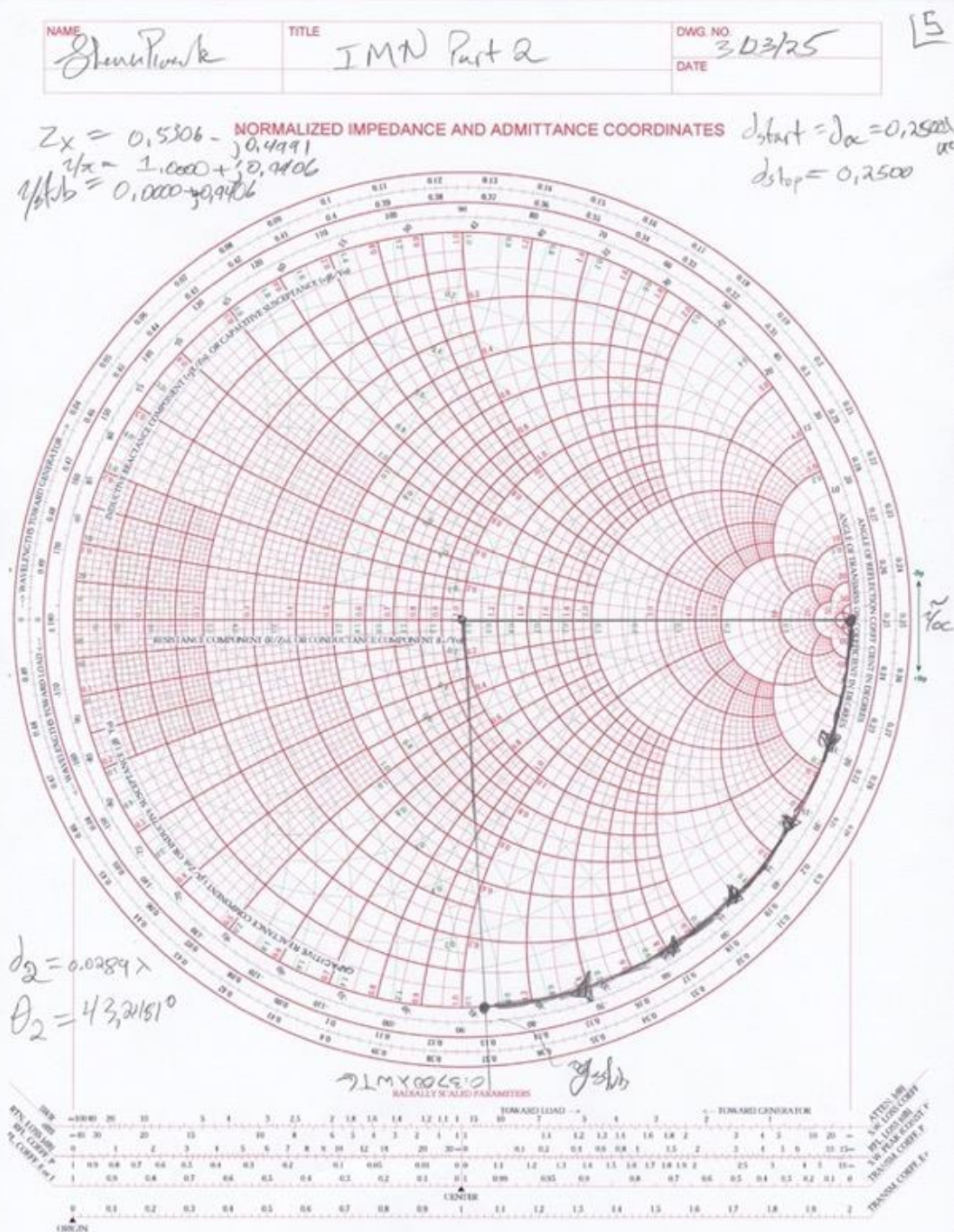
Γ_L - Point 1	(0.3808 , +164.5521°)
Γ_L - Point 2	(0.3940 , -163.4776°)
Γ_L - Point 3	(0.2049 , -143.2555°)
Γ_L - Point 4	(0.1785 , +149.3877°)

Design 2 – Network Layout



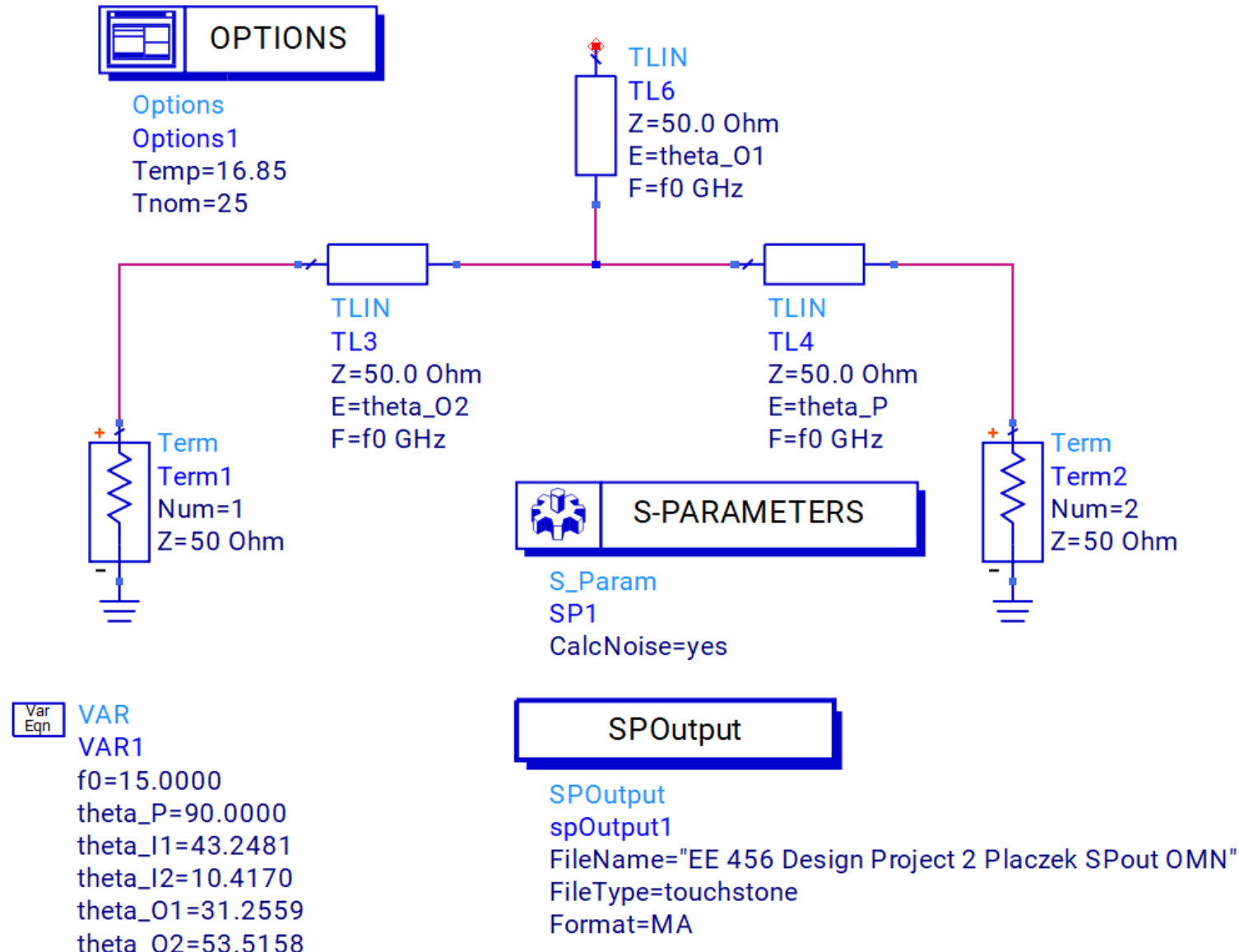
IMN Network Design in ADS

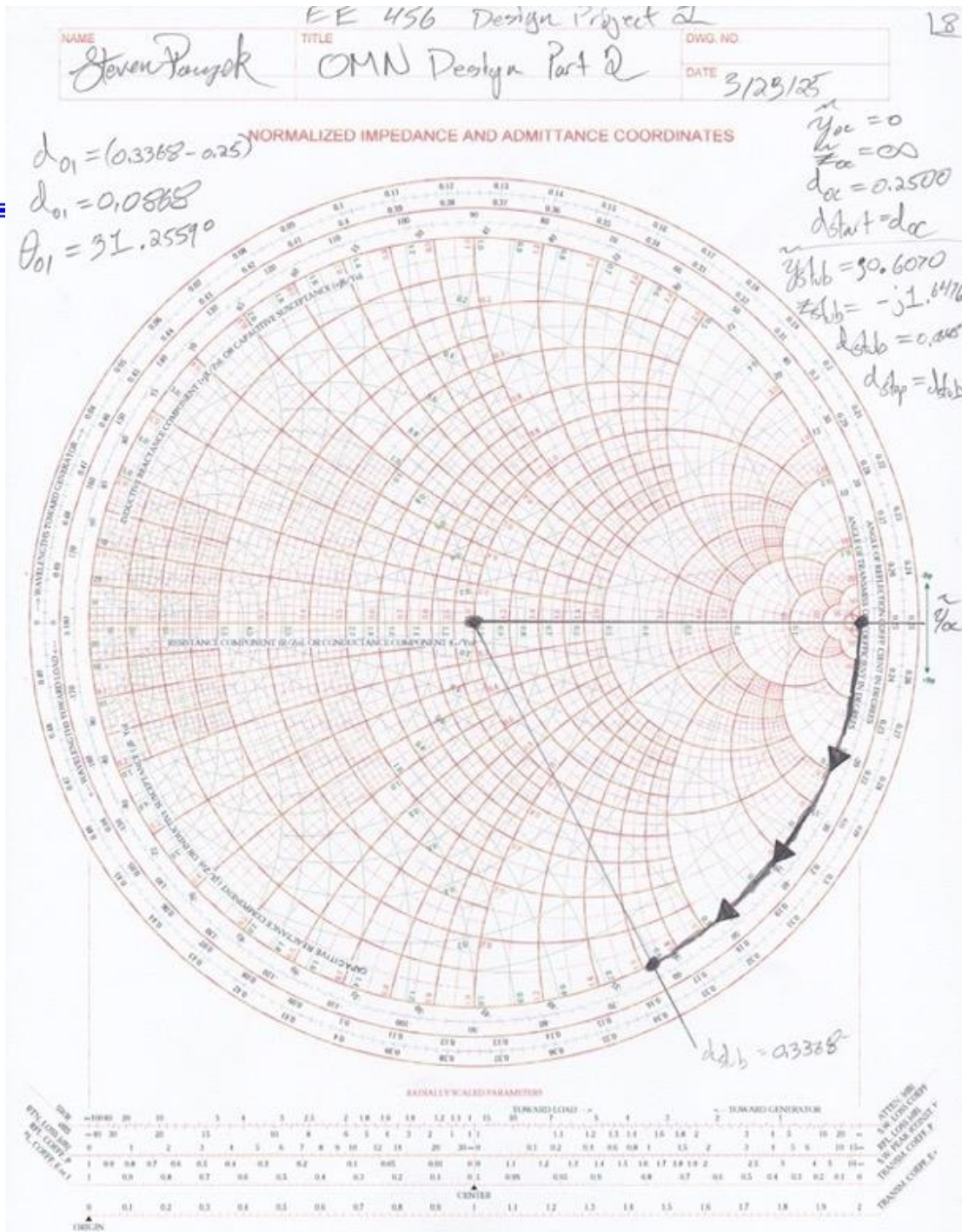




IMN Network Design Part Two

OMN Network Design in ADS





OMN Network Design Part Two

Design 2



Design Parameter	Design 2	Design Spec	Units	Design - Pass / Fail
Γ_S	(0.4256 , -136.0227°)	-----	----	-----
Γ_L	(0.2904 , +146.0865°)	-----	----	-----
Γ_{in}	(0.4742 , +123.6207°)	-----	----	-----
Γ_{out}	(0.3642 , -168.6686°)	-----	----	-----
$ \Gamma_{IMN} $	0.1350	-----	----	-----
$ \Gamma_{IMN} $	-17.3944	-----	dB	-----
$VSWR_{IMN}$	1.4983	≤ 1.5000	----	Pass
$ \Gamma_{OMN} $	0.1630	-----	----	-----
$ \Gamma_{OMN} $	-15.7583	-----	dB	-----
$VSWR_{OMN}$	1.3894	≤ 1.5000	---	Pass
G_T	12.2599	≥ 12.0000	dB	Pass
F_{dB}	0.5803	≤ 0.6000	dB	Pass

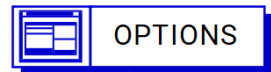
Design 2



IMN Electrical Lengths	λ	Degrees (°)	Network Component	Termination
Theta_I1	0.1201	43.2481	TRL	N/A
Theta_I2	0.0289	10.4170	Shunt Stub	Open

OMN Electrical Lengths	λ	Degrees (°)	Network Component	Termination
Theta_O1	0.0868	31.2559	TRL	N/A
Theta_O2	0.1487	53.5158	Shunt Stub	Open

ADS & MATLAB Simulations



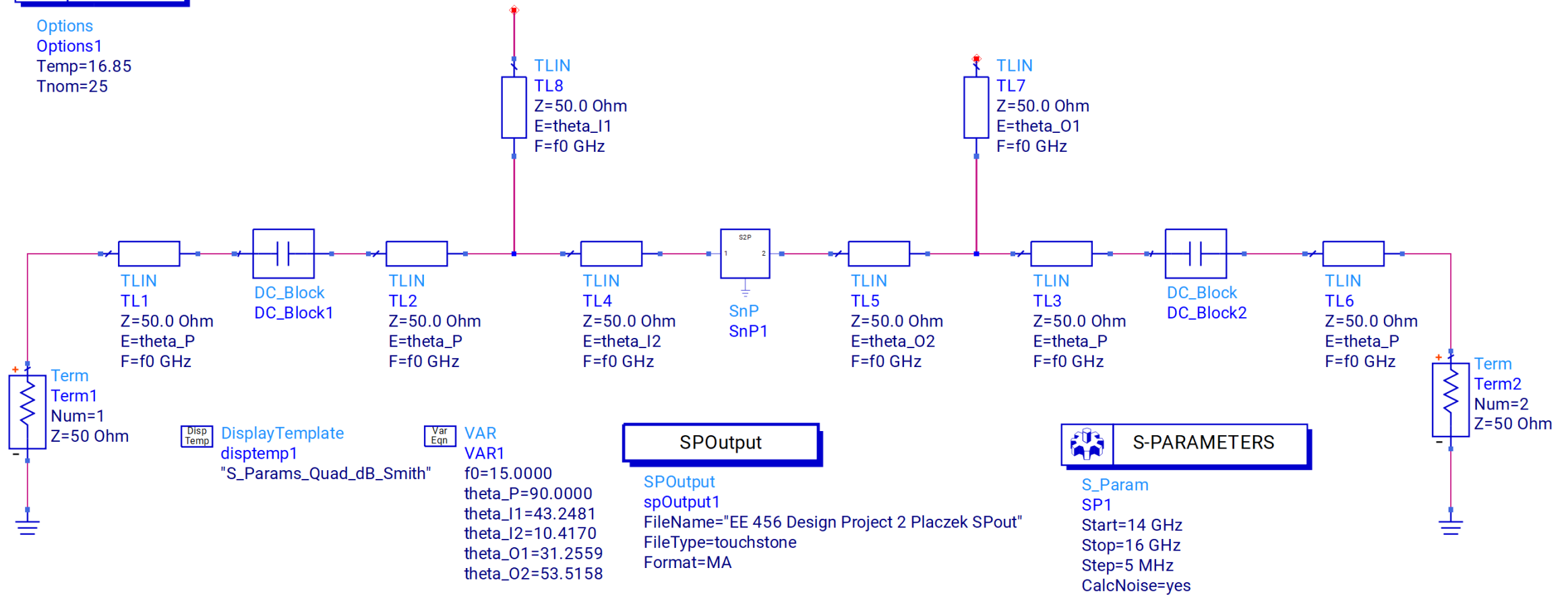
OPTIONS

Options

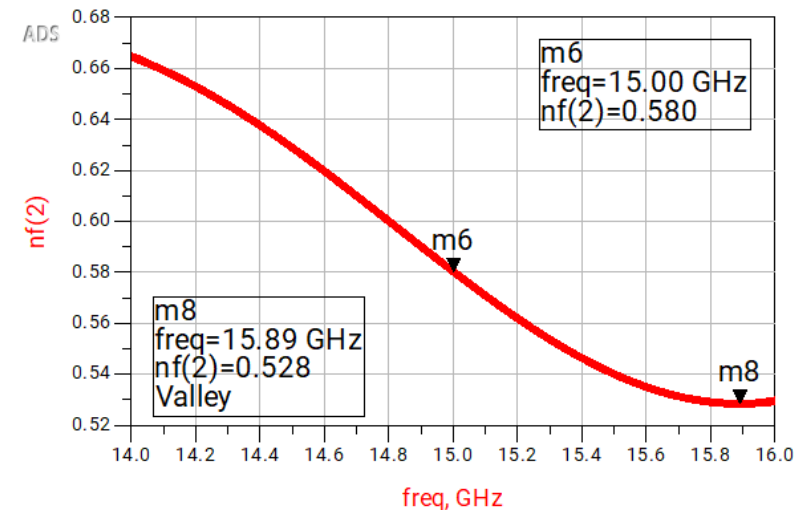
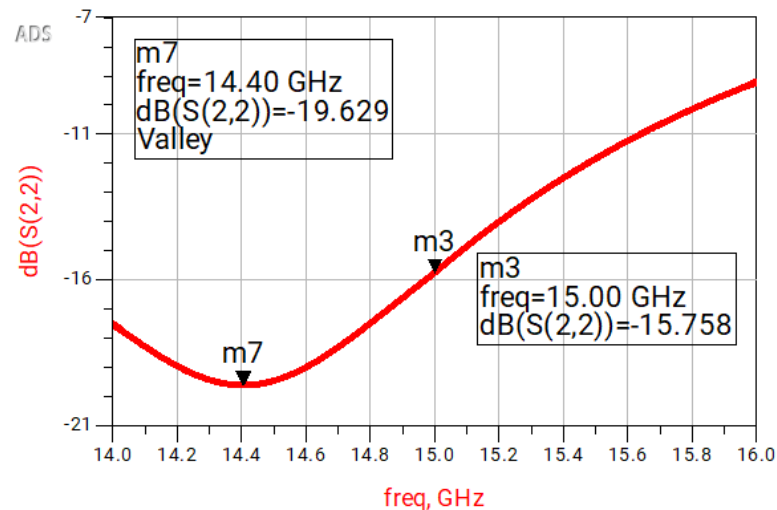
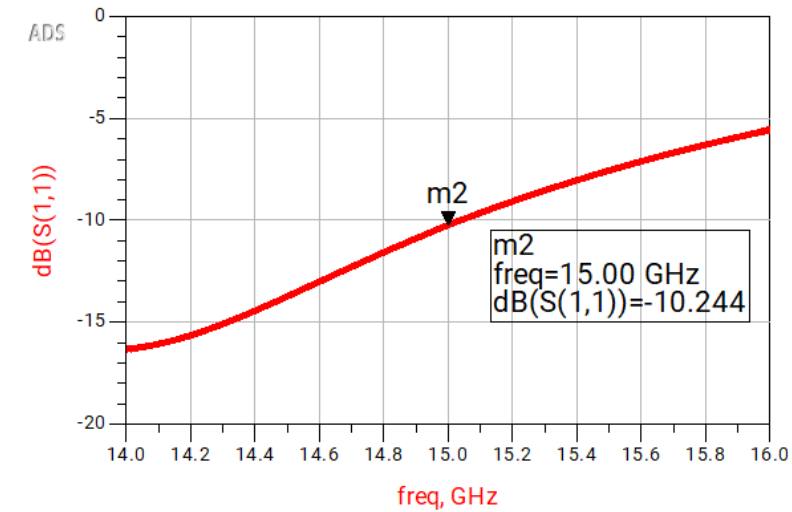
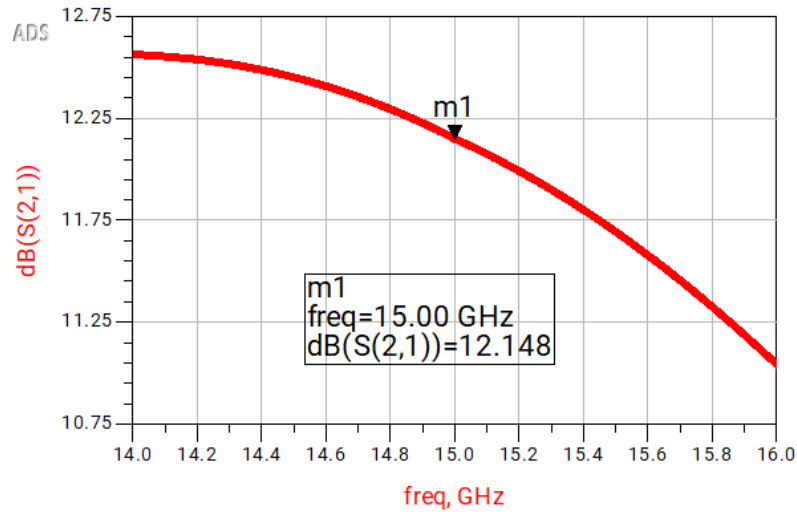
Options1

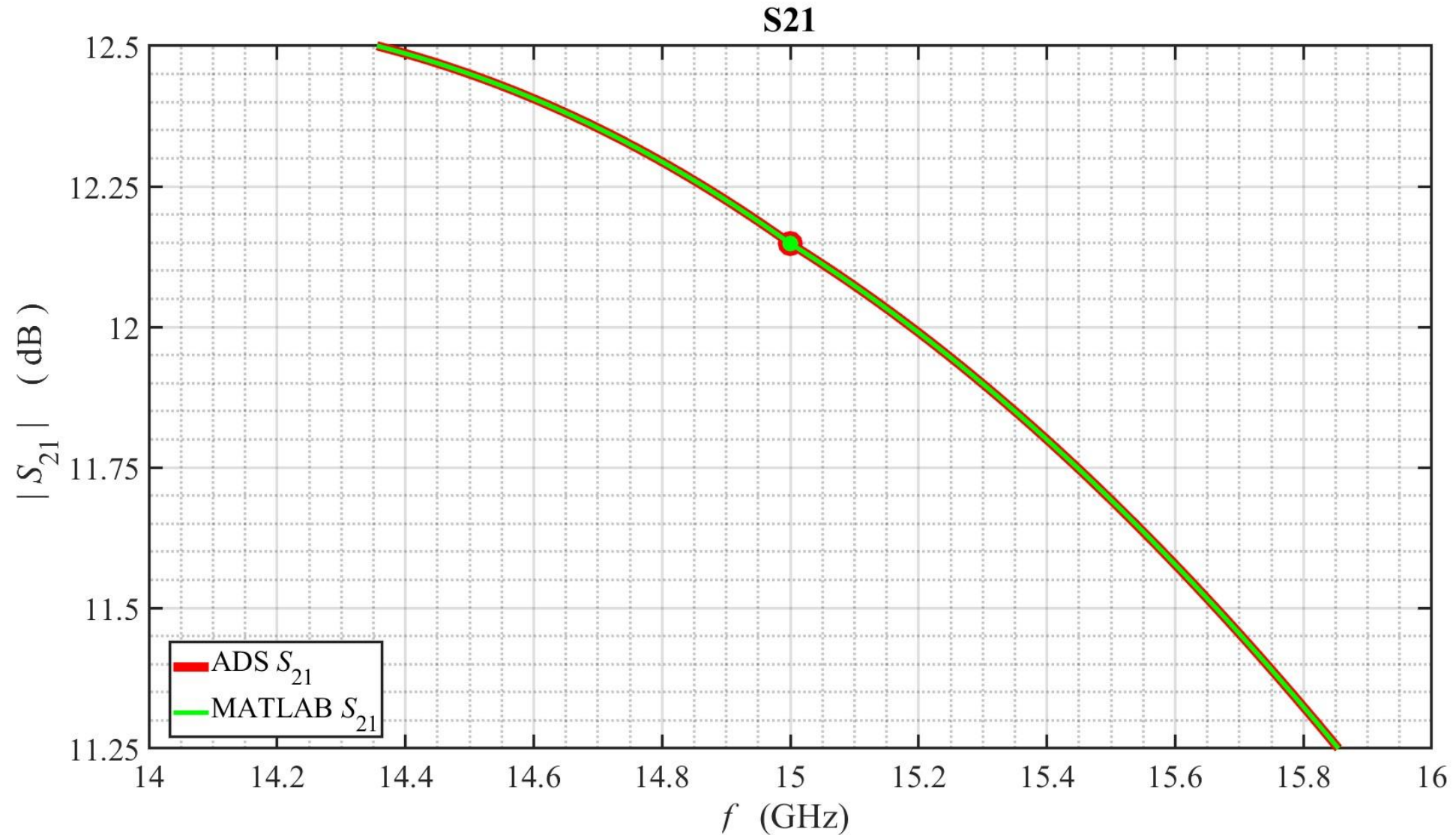
Temp=16.85

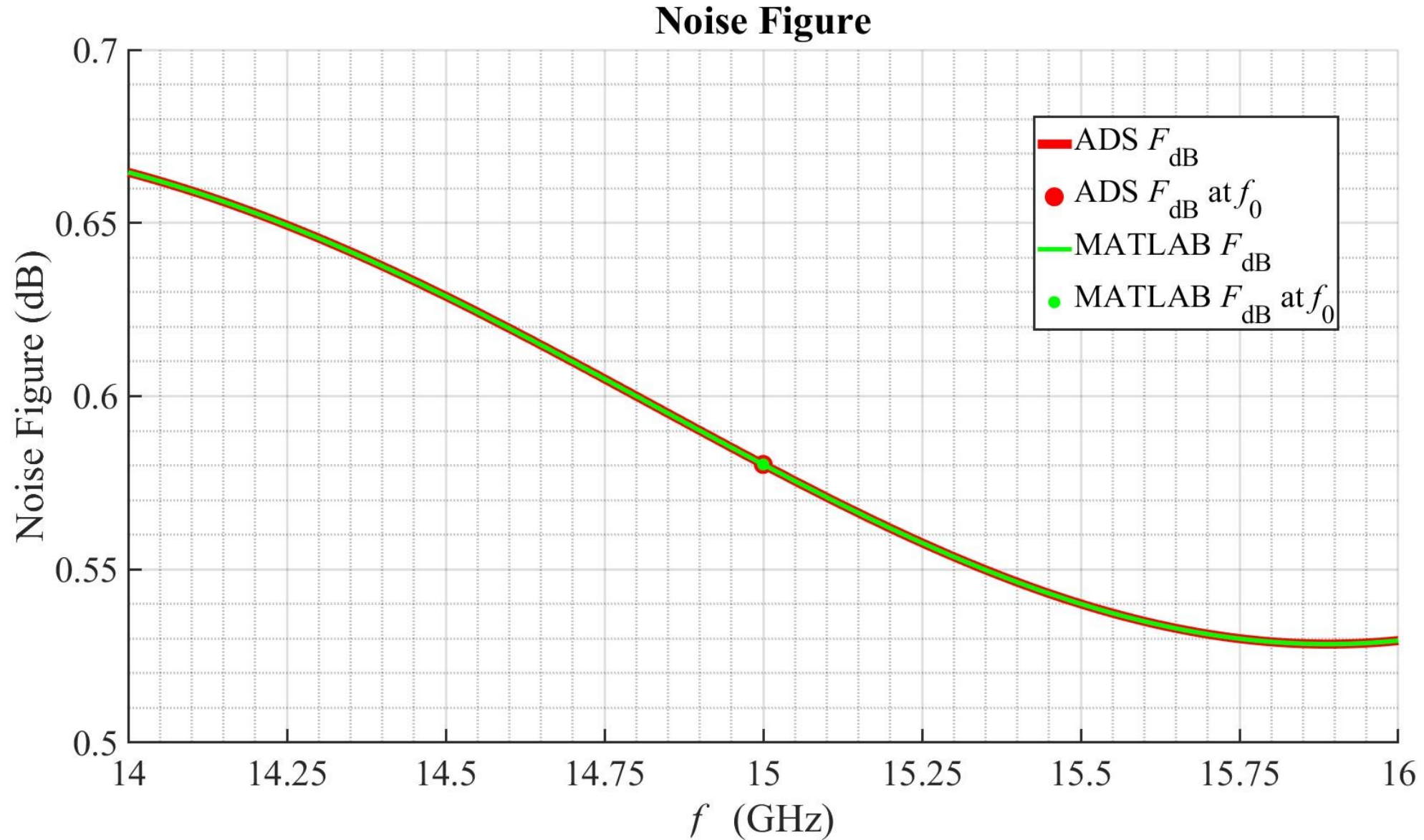
Tnom=25

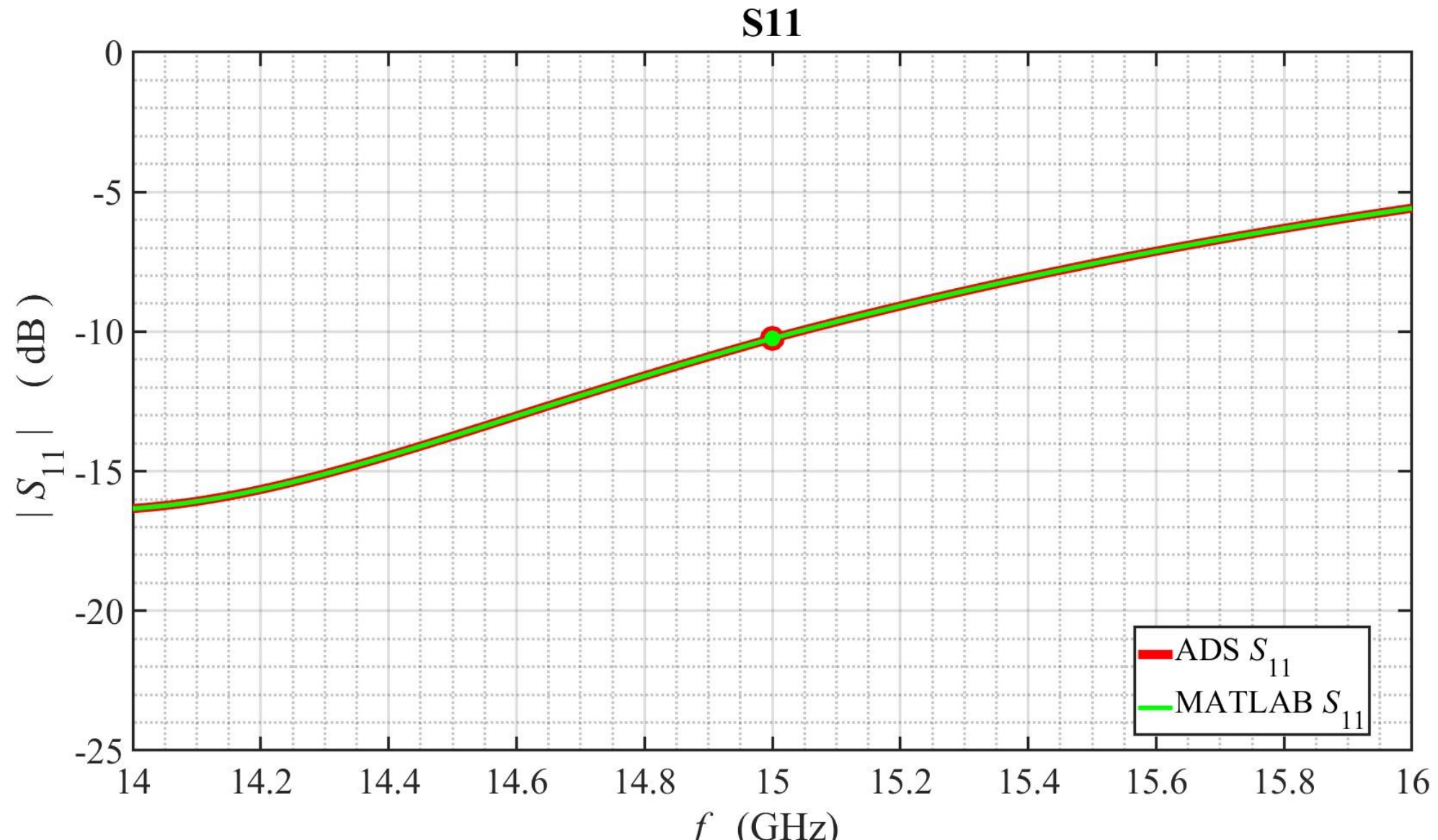


ADS Simulations

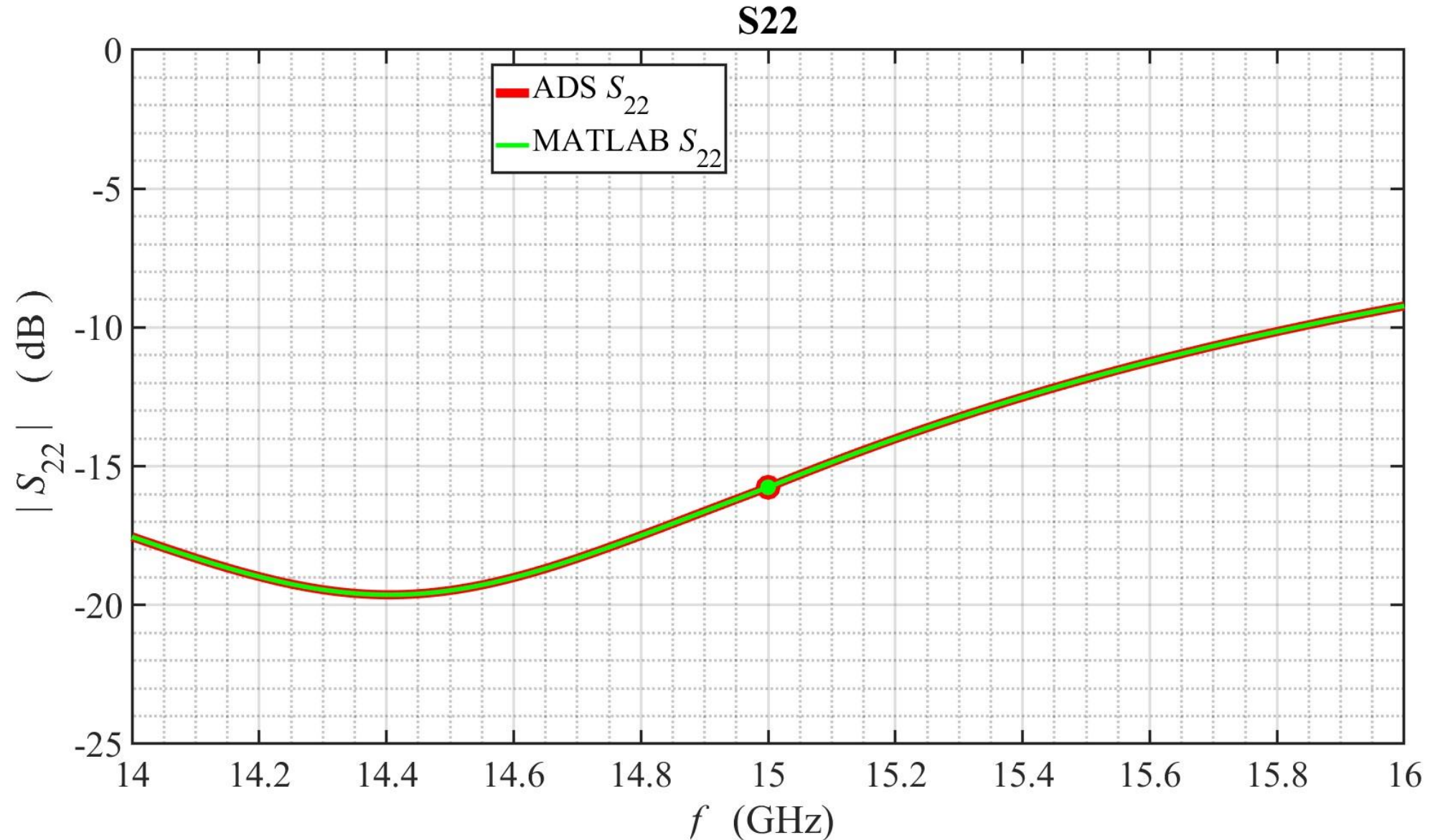


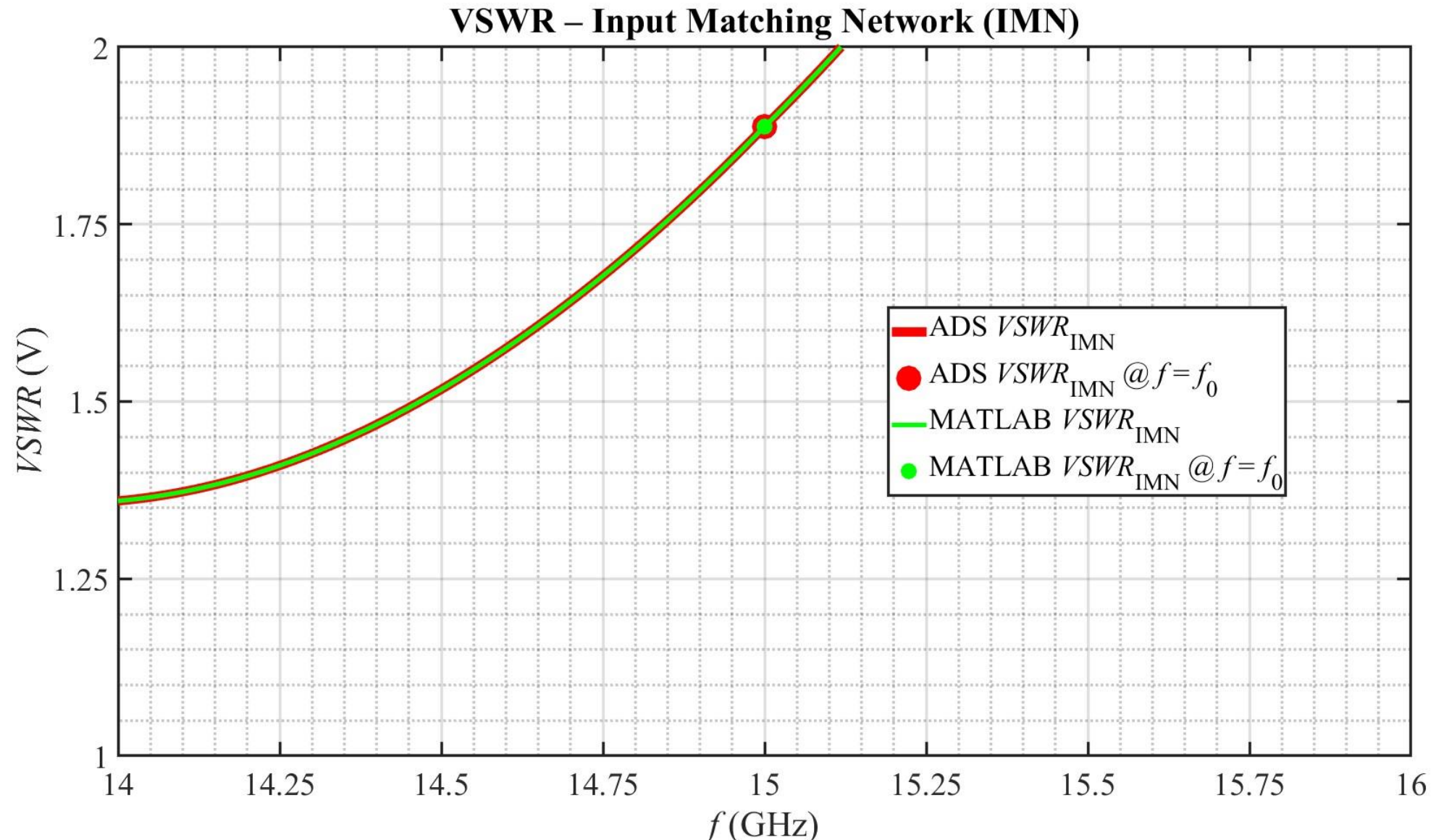


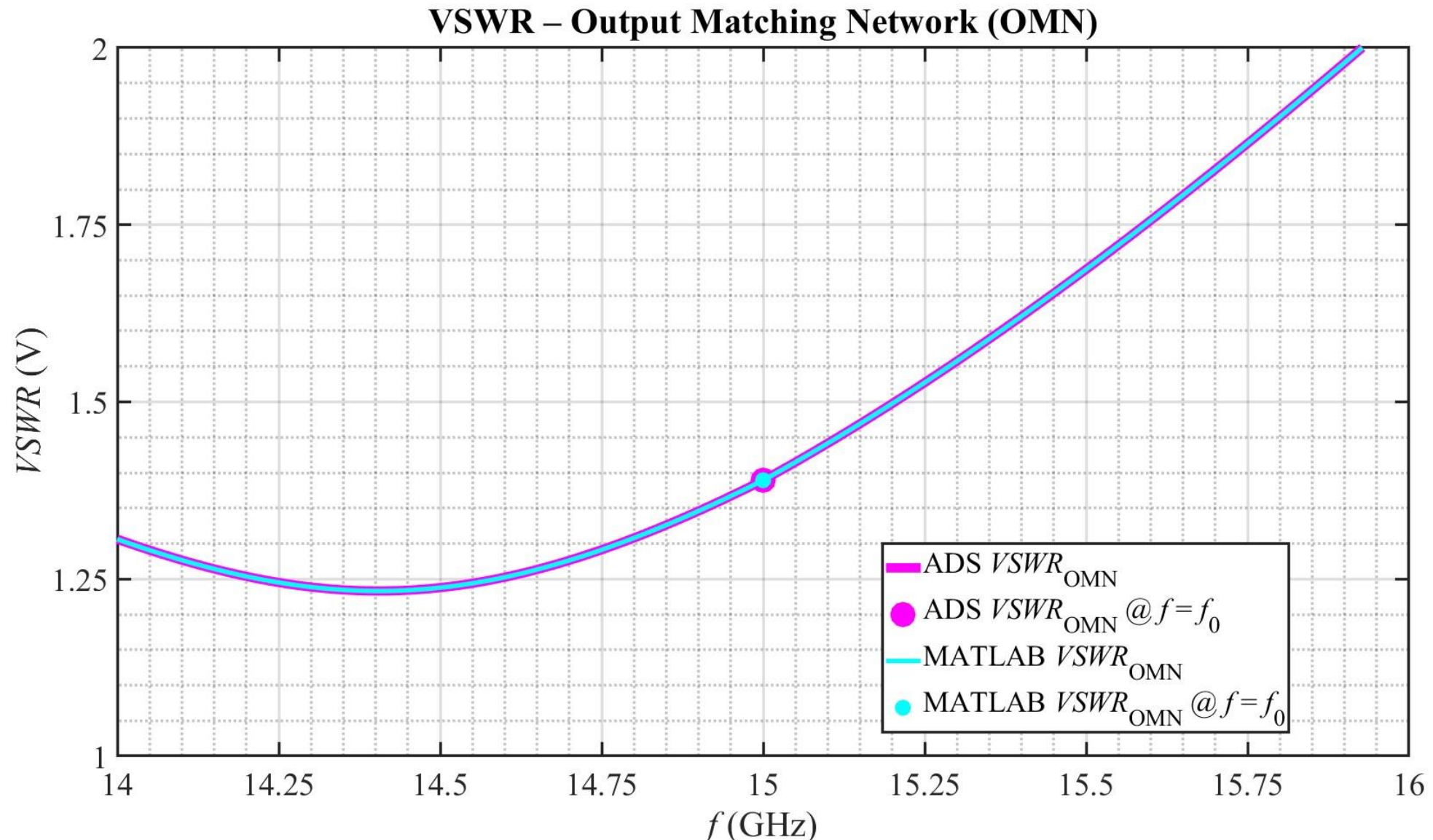




ADS & MATLAB Simulations







Simulation Comparison

This LNA configuration succeeded in meeting all the specifications which the design required.

Both the MATLAB and the ADS simulations provide the same values for the LNA.

MATLAB and ADS provided simulated values for the parameters listed in table 1, these values are essentially the same, showing an excellent match:

- there is less than a 0.0001 percent difference between MATLAB and ADS values.

The noise figure could be lower, however overall gain provided by this design isn't as lacking as many of the other designs solutions which were considered before this one.

End of Design Project 2 Presentation

Additional Material Submitted to Kodiak:

- Excel
- Matlab Code
- ADS Files
- Scans / Screenshots

Appendix I — Excel Spreadsheets

	Design	Simulated Matlab	Simulated ADS	Units		ADS vs Matlab Diff	ADS vs Matlab % Diff
Gamma_S	(0.4256 , -136.0227°)	-----	-----	---		-----	-----
Gamma_L	(0.2904 , +146.0865°)	-----	-----	---		-----	-----
Gamma_In	(0.4742 , +123.6207°)	0.4688 , -10.5628°	(0.5638 , +112.7047°)	---		-----	-----
Gamma_Out	(0.3642 , -168.6686°)	0.3555 , -25.0029°	(0.3642 , -168.6684°)	---		-----	-----
 Gamma_IMN 	0.1350	0.3031	0.3031	---		0.5803	0.0000
 Gamma_IMN 	-17.3944	-10.2445	-10.2445	dB		9.9673	0.0000
VSWR_IMN	1.4983	1.8879	1.8879	---		2.1651	0.0000
 Gamma_OMN 	0.1630	0.1629	0.1629	---		0.4401	0.0000
 Gamma_OMN 	-15.7583	-15.7583	-15.7583	dB		15.4811	0.0000
VSWR_OMN	1.3894	1.3894	1.3894	---		1.6666	0.0000
G_T	12.2599	12.1483	12.1483	dB		12.4255	0.0000
F	0.5803	0.5803	0.5803	dB		0.8575	0.0000

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END